Mohammad Al-Abbas

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EDUCATION

Joint Doctor of Philosophy in Demography & Sociology, University of Pennsylvania**,** Expected: May 2028.

Master of Urban Spatial Analytics, University of Pennsylvania**,** Expected: May 2027

Master of Arts in International Affairs, Lebanese American University**,** May 2023

Bachelor of Engineering in Electrical Engineering, Lebanese American University, August 2019

PUBLICATIONS

1. **Ecological and Socioeconomic Determinants of Childhood Infectious Diseases: The Role of Biogeographic Realms and Ecoregions in Disease Stratification (In Progress)**
   Mohammad Al-Abbas, Emily Hannum, & Jere Behrman. Journal Article.

2. **Ecological Heterogeneity in Malaria Risk and Child Health in Sub-Saharan Africa (In Progress)**
   Mohammad Al-Abbas, Emily Hannum, & Jere Behrman. Journal Article.

3. **Satellite insights and social realities: Stratified vulnerabilities to flooding in Pernambuco, Brazil (Under Review)**
   Mohammad Al-Abbas,* Emily Hannum, Jere Behrman, and Leticia Marteleto. Journal Article. *Population and Environment*

4. **Born in a Haze: Health impacts of early-life exposure to forest fires and PM2.5 in Indonesia (In Progress)**
   Mohammad Al-Abbas, and Emily Hannum. Journal Article.

5. **Environmental Determinants of Migration: Air Pollution in Developing Nations (2022)**
   Mohammad Al-Abbas, Blog Article. *IMS Blog – Borders and Limitations* **Link**

6. **Gender Inequity in Engineering Higher Education: A Case Study of an American University in A Middle Eastern Country (2022)**
   Mohammad Al-Abbas, Conference Article. *2022 10th International Conference on Information and Education Technology.*

7. **The Impact of Faculty Scholarly Collaborations with Non-Doctoral Students (2022)**
   Mohammad Al-Abbas, Conference Article. *IEEE's Interdisciplinary Research in Technology & Management*

8. **Quantifying Publisher's Competence Through Scholarly Engagement (2021)**
   Mohammad Al-Abbas, Samer S. Saab, Journal Article. *Publishing Research Quarterly*

9. **A Case Study of Gender Diversity in a Middle Eastern University (2021)**
Jamil Ahmad Haidar, Joelle Oneissi, & **Mohammad Al-Abbas,** Conference Article**.** *19th International Conference on Information Technology Based Higher Education and Training*

10. **Setting the Boundaries of COVID-19 Lockdown Relaxation Measures (2021)**
Samer S. Saab, **Mohammad Al-Abbas**, Roula Diab, Rayana Jaafar, Khaled S. Saab, & Samer S. Saab, jr. Journal Article. *Library Hi Tech*

11. **Sustainable Gender Equity Influenced by an Americanized Lebanese Pedagogy (2021)**
Joelle Oneissi, **Mohammad Al-Abbas**, Conference Article. *Proceedings of the World Conference on Women's Studies*

12. **Toward an Improvement of Engineering Teaming Skills Through an In-House Professionalism Course (2020)**
Mohammad Al-Abbas, Wissam Fawaz, Raymond Ghajar, Chadi Abou Rjeily, Iyad Ouaiss, Zahi Nakad, George E. Nasr, & Samer S. Saab, Journal Article. *IEEE Transactions on Education*

13. **Formation of Policies Guided by Multivariable Control Theory (2020)**
Joe J. Khalife, **Mohammad Al-Abbas**, & Samer S. Saab, Journal Article. *Operations Research Perspectives*

14. **The Impact of Collaborative Research-A Case Study in a Developing Country (2020)**
Mohammad Al-Abbas, Samer S. Saab, Conference Article. *IEEE Symposium on Multidisciplinary Studies and Innovative Technologies*

15. **On the Impact of Multi-Authorship Scholarly Publications (2020)**
Mohammad Al-Abbas, Samer S. Saab, Conference Article. *IEEE International Conference on Advent Trends in Multidisciplinary Research and Innovation*

TALKS & GUEST SPEAKER

OCT 2025
Guest Lecturer on Fertility Transition, University of Pennsylvania – Population Studies Center
Delivered a doctoral lecture in Introduction to Demography on fertility dynamics in low- and middle-income countries and the emergence of sub-replacement fertility in high-income countries.

AUG 2025
Conference Presenter, American Sociological Association Annual
Presented research on Floods in Brazil and the stratified, gendered impacts on women.

AWARDS & CERTIFICATIONS

Nov 2023
Social and Behavioral Responsible Conduct of Research, CITI Program
RCR covers core norms, principles, regulations, and rules governing the practice of research. Granted as part of my Ph.D. in Demography.

Nov 2023
Social and Behavioral Research, CITI Program
Basics of Human Research. Granted as part of my Ph.D. in Demography.

APR 2022
Best Presentation Award, ICIET 2022 – Session 2
Granted for my performance at the 10th International Conference on Information and Education Technology at Matsue, Japan.

JULY 2021
Outstanding Researcher Award, Lebanese American University
Granted for outstanding scholarly performance in the subject area of social sciences and management.

APR 2018
Protecting Human Research Participants, NIH Office of Extramural Research
Granted for my competition of NIH's training course as part of my duties towards the IRB committee

EXPERIENCE

August 2025
Teaching Assistant, University of Pennsylvania, Graduate School of Education (GSE) — Philadelphia, PA
- Led weekly quantitative lab sections introducing core statistical concepts for program evaluation, with emphasis on regression-based inference and interpretation.
- Developed and delivered hands-on tutorials in Stata, including reproducible code templates, annotated walkthroughs, and troubleshooting guides to support students with varied quantitative backgrounds.
- Supported students' applied evaluation projects by advising on research design, variable construction, model specification, and communicating results with appropriate assumptions and limitations.
- Held regular office hours and provided individualized feedback on assignments and lab deliverables, focusing on translating evaluation questions into defensible analytic strategies.
- Coordinated with the instructor to align lab content with lecture topics, grading rubrics, and learning objectives.

OCT 2023
Research Assistant, University of Pennsylvania, PIRE Project
- Under the supervision of Professors Emily Hannum & Jere Behrman;
- An NSF funded project that tackles the effects of environmental degradation on child development in low-to middle income countries;
- Current Output Paper: Born in a Haze: Health impacts of early-life exposure to forest fires. Which aims to improve current understanding forest fires on birth outcomes in Indonesia. Provides advancements on forest fires clustering, fixed bandwidth approximations, and k-means quintile regression algorithms;
- Assisting Prof. Heather and Prof. Leticia on their respective statistical needs.

OCT 2023
Statistical Consultant, University of Pennsylvania, Department of Sociology
- Provide statistical support for PhD students in Sociology and Demography;
- Curate databases for PhD students; and
- Conduct workshops on Machine Learning Techniques

SEP 2022 – MAR 2023
Investigations Section – Intern, United Nations Development Programme (UNDP) Office of Audit and Investigations
- Assist in reviewing cases reported to OAI;
- Assist in drafting documents and communications with respect to cases under investigation, including investigation plans, witness statements, investigation reports, closure reports and other documents;
- Assist in conducting interviews with subjects, complainants or witnesses;

- Assist in identifying, collating and analyzing potential evidence, including carrying out open-source research and forensic reviews; and
- Assist with the editing and review of investigation reports and other documents to ensure that citations to evidence are accurate and complete.

SEP 2022 – JAN 2023
Technology Policy Program – Intern, **Center for International Policy**
- Design the Social Media Harms tracker for CIP;
- Locate data and news on social media harms within the MENA, African, LATAM, and SEA regions;
- Author policy briefs and reports on time-sensitive matters;
- Attend legislative hearings in both the US Senate and House of Representatives; and
- Maintain an internal Hill tracker to keep the program up-to-date with current legislative news

JAN 2022 – AUG 2022
Research Assistant – Individual Contractor, United Nations Economic and Social Commission for Western Asia (ESCWA)
- Provided statistical support in the form of imputations, machine learning algorithms, and coding;
- Illustrated statistical data;
- Responsible for filling and maintain the social demographic database of ESCWA, with over 22 countries and 300 indicators; and
- Demonstrated excellent project completion skills. At the time of the contracts completion, ESCWA's social demographic base's completion rate rose from 29% to 83%.

JUNE 2019 – DEC 2021
Monitoring and Evaluation Officer, Lebanese American University – Office of Graduate Studies and Research
- Supported faculty in their research activities (e.g., journal selection);
- Provided bibliometric analysis of scholarly journals;
- Proposed research cohorts/collaborations between faculty, staff, and students based on previous published works; and
- Monitored the performance of faculty, departments, and schools across various KPIs and databases.

AUG 2019 – DEC 2021
Graduate Teaching Assistant, Lebanese American University – Department of Electrical and Computer Engineering
- Advised students on coursework;
- Taught Computer Proficiency, Engineering Programming, Engineering Graphics, and Microprocessor's labs; and
- Held and proctored exams.

AUG 2019 – AUG 2020
Non-voting IRB Member, Lebanese American University – Institutional Review Board
- Reviewed consent forms, questionnaires, and research methodologies

SEP 2018 – JAN 2019
USP Mentor, Lebanese American University – USAID
- Aided USAID grant students with their academic endeavors;
- Ensured Mentee's adherence to university and USAID's Rules & Regulations;
- Coordinated between the students and the grant office;
- Tracked achievements and grades of mentees; and
- Designed progress reports for each mentee.

JUN 2015 – JUN 2019
Student Research Assistant, Lebanese American University – Department of Electrical and Computer Engineering

- Collected data through interviews and qualitative surveys;
- Conducted literature reviews on various topics to identify new avenues for research; and
- Prepared team meeting minutes and outline research agendas.

FELLOWSHIPS & SCHOLARSHIPS

AUG 2024 – MAY 2028
Benjamin Franklin Fellowship, Issued by Graduate Group in Demography – University of Pennsylvania
A fellowship awarded upon entry and provides a stipend, healthcare waiver, and a tuition release for four years.

AUG 2023 – JUL 2024
Dean's Fellowship, Issued by the Office of the Dean of School of Arts & Sciences – University of Pennsylvania
An additional fellowship that provides one extra year of funding, and has covered my first year of studies at the institution.

SEPT 2021 – MAY 2023
MEPI – Tomorrow's Leaders Graduate Program, Lebanese American University
Tomorrow's Leaders Graduate (TLG) program is a MEPI initiative aiming at offering students from the Middle East and North Africa (MENA) a unique opportunity to pursue higher education and earn a master's degree at a leading American University with U.S. accreditation in the MENA region. TLG equips graduate students with the professional experience needed to advance knowledge and contribute to the achievement of socially relevant outcomes through research that furthers the professionalization of emerging leaders. In addition, the program increases scholars' knowledge of gender issues in the various fields of study.

SEPT 2021
Research Affiliate, Institute for Migration Studies at Lebanese American University
The Institute for Migration Studies (IMS) at the School of Arts and Sciences at LAU is an interdisciplinary, policy-driven and network-oriented research institute that develops research in the areas of Migration, Refugee and Displacement Studies. IMS serves as a resource center for graduate students, researchers, activists, policy analysts, humanitarian staff and scholars interested in the study of migration and its various impacts on Lebanon and the MENA region. In line with the overall mission of LAU, IMS aims to advance new narratives and intersectional approaches to the study of migration through promoting research, generating studies, publishing the IMS' working paper series and academic blog, and organizing public webinars, conferences and workshops as well as through its membership in various regional and international networks.

SKILLS

- **Programming:** R, Python, STATA, JAVA, and Assembly
- **Machine Learning:** Natural Language Processing (NLP), Reinforcement Learning (RL), Neural Networks (CNN & RNN), and Decision Trees.
- **Software:** R Studio, STATA, ArcGIS, Git, SLURM, LaTeX (Overleaf), and Tableau

LANGUAGES

- English (Bilingual Proficiency)
- Arabic (Native Proficiency)

VOLUNTEER WORK

- URDA – Teaching Initiative
- Beirut for the Ethical Treatment of Animals

- Engineers Without Borders (EWB)
- FoodBless – Feeding the Hungry